

Student Name _____

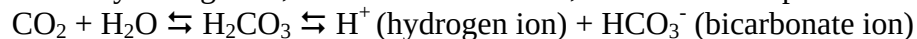
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ACTIVITY 20: Investigating CO₂ in breathing (full write-up)

Introduction:

In this laboratory session you will be presented with several problems that will allow you to investigate various aspects of breathing. You will be expected to devise experimental procedures to investigate these problems and seek their solutions.

When your cells use food for energy, the waste products that are produced are carbon dioxide and water. These compounds may be found in the circulatory system until they eventually leave the body. CO₂ is carried in the blood in three ways, as carboxyhemoglobin, as the bicarbonate ion, or dissolved in plasma.



It should be noted that when CO₂ reacts with water, the pH of the water is lowered to a point between pH 5 and pH 6 (acidic).

Materials:

paper bags	twin bottle respirometer	phenolphthalein	bromthymol blue
test tube rack	straws	beaker	4 large test tubes
aquarium beaker	fish nets with fish	dialysis bag	seltzer water
lime water	graduated cylinder	paper towels	

Notes on the Reagents:

- 1) Phenolphthalein is a pH indicator: pink above a pH of 8.3, colorless below.
- 2) Bromthymol blue is yellow at pH 6 and below. It becomes green above 6 and blue at pH 7.6 and above.
- 3) Seltzer water is a saturated solution CO₂
- 4) Limewater is a clear aqueous solution of Ca (OH)₂ (calcium hydroxide). When CO₂ reacts with limewater it produces a milky suspension from which a chalky solid will precipitate.

Respirometer Use:

Study the diagram below. Observe the arrangement of tubes and trace the path of airflow by following the arrows. Note that normal breathing through the mouthpiece will bring fresh air in to you through chamber A as you breathe in, and that as you exhale, your exhaled air will pass through chamber B. Be sure that you breathe only through your mouth directly into the mouthpiece. Keep your breaths regular and slow. You cannot breathe in any phenolphthalein, so don't be nervous!

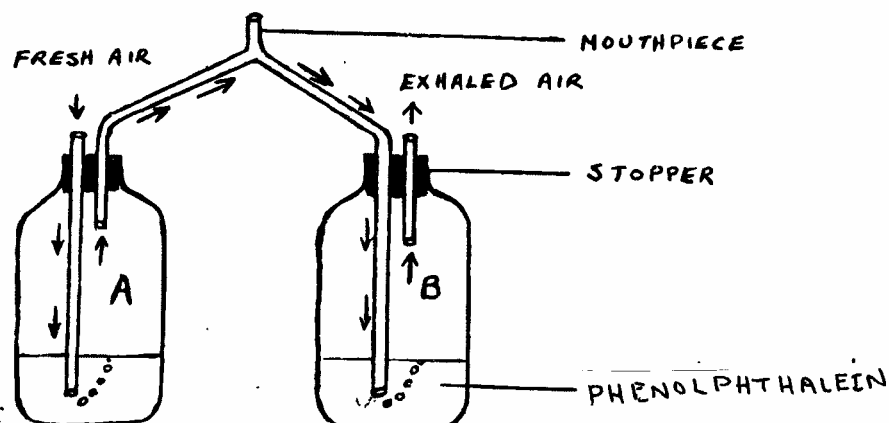


Figure 1. The Respirometer

Problems to Solve: Write an experimental procedure to help answer the following questions. Record the results for each in a separate table.

- 1) How does exhaled air at rest, compare with inhaled air with respect to its carbon dioxide content? (To answer this question, assume the air in the room has a carbon dioxide content of 0.05%) **Hint: use your twin-bottle respirometer.** Enter the results of this experiment in Table 1 and on the board.
- 2) How does the percentage of CO₂ in exhaled air after exercise compare to the percentage of CO₂ in exhaled air at rest?
- 3) Compare pulse rates at rest and immediately after exercise. How does the change in pulse rate correlate with changing breathing rate?
- 4) How many breaths per minute does a person take at rest compared to the number of breaths per minute immediately after exercise? (Exercise: running in place for 2 minutes with high knee movements.)
- 5) What happens to the breathing rate if the CO₂ concentration of inhaled air is increased?
- 6) How can we determine if goldfish produce CO₂ when they breathe?
- 7) How can we determine if CO₂ is readily diffusible through a membrane?

Table 1.

Group number	Breaths at rest to turn phenolphthalein colorless (exhaled air)		Breaths at rest to turn phenolphthalein colorless (inhaled air)		Breaths after exercise to turn phenolphthalein colorless (exhaled air)		Breaths after exercise to turn phenolphthalein colorless (inhaled air)	
	Breaths	% Carbon dioxide	Breaths	% Carbon dioxide	Breaths	% Carbon dioxide	Breaths	% Carbon dioxide
1				0.05%				0.05%
2				0.05%				0.05%
3				0.05%				0.05%
4				0.05%				0.05%
5				0.05%				0.05%
6				0.05%				0.05%
7				0.05%				0.05%
8				0.05%				0.05%
9				0.05%				0.05%
average				0.05%				0.05%

Table 2.

Group number	Pulse rate at rest	Pulse rate after exercise	Breathing rate at rest	Breathing rate after exercise
1				
2				
3				
4				
5				
6				
7				
8				
9				
Average				

Table 3. Results of increasing the CO₂ content in the air we breath

Group #	Before Paper Bag		After Paper Bag	
	Breaths/minute	Pulse/minute	Breaths/minute	Pulse/minute
1				
2				
3				
4				
5				
6				
7				
8				
9				
Average				

Table 4. Ability of CO₂ to diffuse through a membrane

Group #	Reaction inside dialysis membrane	Reaction outside dialysis membrane
1		
2		
3		
4		
5		
6		
7		
8		
9		

Discussion:

- 1) Using the mean data from the twin-bottle respirometer experiment, compute the carbon dioxide content of exhaled air.
- 2) Compare the direction of diffusion of CO₂ in the lungs and in all other body tissues.
- 3) Draw a graph showing the effect that an increase in the CO₂ level in the blood has on the rate of breathing.
- 4) Why did breathing into a bag simulate the effect of physical exercise?
- 5) Why is it reasonable to expect that there is a direct relationship between breathing rate and pulse rate?